

Nathan Tebbs

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SKILLS

Languages: C/C++, Java, Python, Go, Rust, Odin, JavaScript, Lua, SQL, Bash

Tools: Git/GitHub, Docker, Linux, systemd, GDB, Valgrind, SSH, VMware/KVM, Neovim & Emacs

Libraries: SDL3, Raylib, HLSL, pandas, NumPy, scikit-learn, Streamlit

EDUCATION

University of Arizona

Bachelor of Science in Computer Science

• Relevant Coursework: Compilers, Database Design, Algorithms, Computer Networking, Cloud Computing, Systems Programming

South Puget Sound Community College

Associate of Arts

Tucson, Arizona

August 2023 — May 2026

Olympia, Washington

September 2021 — May 2023

WORK EXPERIENCE

Software Engineer (Web)

Sound Psychiatric Services

January 2021 — Present

Olympia, Washington

- Deployed and maintained production websites, handling hosting, updates, backups, and long-term maintenance.
- Migrated legacy systems to modern stacks with improved reliability and security posture.

Undergraduate Teaching Assistant (CSc 346 Cloud Computing)

University of Arizona

January 2025 — May 2025

Tucson, Arizona

- Assisted students with Linux-based cloud VMs, web servers, networking, and deployment issues.
- Troubleshoot configuration errors and documented common failure modes and resolutions.

PROJECTS

Author, Mach Game Engine (github.com/vetr0s/mach)

June 2026 — Present

- Building a 2D and 3D game engine in pure C with SDL3, with isometric and standard camera systems and cross-platform shaders.
- Split the engine from the game so the engine runs game code through a callback interface and never depends on game types, keeping it reusable across projects.
- Wrote the core subsystems from scratch, including vector and matrix math, an SDL_GPU renderer, input handling, and the main game loop, all built as a unity build with one `clang` call and no separate build system.

Author, GateRelay: Secure Linux TCP Relay Service (github.com/vetr0s/gaterelay)

January 2026 — February 2026

- Built a production-grade TCP relay in Go forwarding public ingress traffic to a controlled backend, with connection limits, timeouts, graceful shutdown, and structured JSON logging.
- Hardened it with defense-in-depth: a non-root systemd user, sandboxing (capability dropping, syscall filtering, ProtectSystem), ufw firewall rules, and SSH key-only access, with a documented threat model.

Collaborator, Urban Crime Data Analysis (github.com/vetr0s/tucson-crime-analysis)

March 2025 — May 2025

- Ran a multivariate analysis of Tucson Police crime data against neighborhood income and streetlight locations to test how socioeconomic and environmental factors relate to crime.
- Built preprocessing, geospatial, and modeling pipelines in Python (pandas, geopandas, scikit-learn, statsmodels) using Ridge Regression, Random Forest, Logistic Regression, and OLS to predict high-crime wards.

EXTRACURRICULAR ACTIVITIES

FIRST Robotics Competition team 4450 (Safety Captain)

September 2021 — June 2023

- Led safety operations for a competitive engineering team, enforcing shop, electrical, and tooling safety standards under strict build and competition deadlines.
- Co-planned an international outreach program, coordinating risk mitigation for a \$60,000 fundraiser that sent 17 students and metal-based FTC robotics kits to Cairo, Egypt for hands-on robotics education.

Google Developer Student Club

August 2023 — May 2026

- Participated in peer-led workshops and collaborative projects on modern software development and Google Cloud technologies.

HackAZ 2026

April 2026 — April 2026

- Built and deployed GridWise Energy with 3 teammates in a 24-hour hackathon, a live web app analyzing energy usage from local power-grid data.
- Shipped authentication, time-series usage tracking, AI insights via a chatbot and an ML forecast model, and an opt-in global leaderboard.